

HDOC - Day 8 Activity

C Programming Assignment

Question 1:

Write a C program to find the radius, diameter, and volume of a sphere when its surface area is given.

Input, Output and Formula

Input: Surface area of the sphere (**A**)

Output: Radius (**r**) ,Diameter (**d**) & Volume (**V**)

Formula Used:

$$r = \sqrt{\frac{A}{4\pi}}$$
$$d = 2r$$
$$V = \frac{4}{3} \pi r^3$$

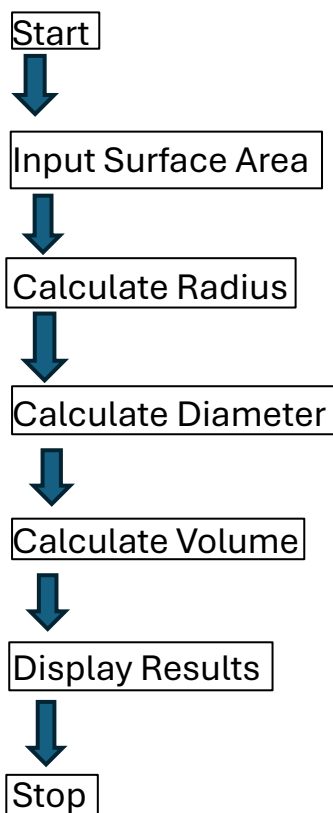
Standard Test Case Table

Test case	Input(A)	Expected output(r, d, V)	Actual output	Result
1	314.16	5,10 ,523.60	Same	Pass
2	50.27	2 ,4 ,33.51	Same	Pass
3	12.57	1 ,2, 4.19	same	Pass

Algorithm

1. Start.
2. Read the surface area.
3. Calculate the radius using the formula.
4. Calculate the diameter.
5. Calculate the volume.
6. Display the radius, diameter, and volume.
7. Stop.

Flowchart



C Program

```
#include <stdio.h>
#include <math.h>
#define PI 3.14159265359
int main() {
    float area, r, d, v;
    printf("Enter surface area: ");
    scanf("%f", &area);
```

```

r = sqrt(area / (4 * PI));
d = 2 * r;
v = (4.0 / 3.0) * PI * r * r * r;
printf("Radius = %.2f\n", r);
printf("Diameter = %.2f\n", d);
printf("Volume = %.2f\n", v);
return 0;
}

```

Compilation , Execution & Output

```

(kali㉿kali)-[~]
$ cc sp.c -lm
linux-gnu-ld.bfd: /tmp/cc1QrQEE.o: in function 'main':
sp.c:(.text+0x15): undefined reference to 'sqrt'
collect2: error: ld returned 1 exit status
(kali㉿kali)-[~]
$ ./a.out
Enter surface area: 314.16
Radius = 5.00
Diameter = 10.00
Volume = 523.60

(kali㉿kali)-[~]
$ ./a.out
Enter surface area: 50.27
Radius = 2.00
Diameter = 4.00
Volume = 33.51

(kali㉿kali)-[~]
$ ./a.out
Enter surface area: 12.57
Radius = 1.00
Diameter = 2.00
Volume = 4.19

```

Question 2:

Write a C program to find the image of a point after reflection in the origin.

Input, Output and Formula

Input: x-coordinate & y-coordinate

Output: Reflected point

Formula Used: Reflection about the origin: $(x,y) \rightarrow (-x,-y)$

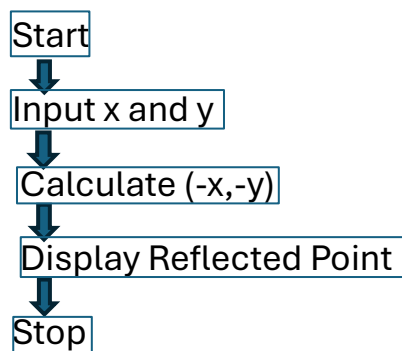
Standard Test Case Table

Test case	Input	Expected output	Actual output	Result
1	(2,3)	(-2,-3)	Same	Pass
2	(-4,5)	(4,-5)	Same	Pass
3	(0,-7)	(0,7)	same	Pass

Algorithm

1. Start.
2. Read x and y.
3. Multiply both coordinates by -1.
4. Display the reflected point.
5. Stop.

Flowchart



C Program

```
#include <stdio.h>
int main() {
    int x, y;
    printf("Enter x and y: ");
    scanf("%d %d", &x, &y);
    printf("Reflected point = (%d, %d)\n", -x, -y);
    return 0;
}
```

Output

```
(kali㉿kali)-[~]  
$ cc image.c  
  
(kali㉿kali)-[~]  
$ ./a.out  
Enter x and y: 2 3  
Reflected point = (-2, -3)  
  
(kali㉿kali)-[~]  
$ ./a.out  
Enter x and y: -4 5  
Reflected point = (4, -5)  
  
(kali㉿kali)-[~]  
$ ./a.out  
Enter x and y: 0 -7  
Reflected point = (0, 7)
```